

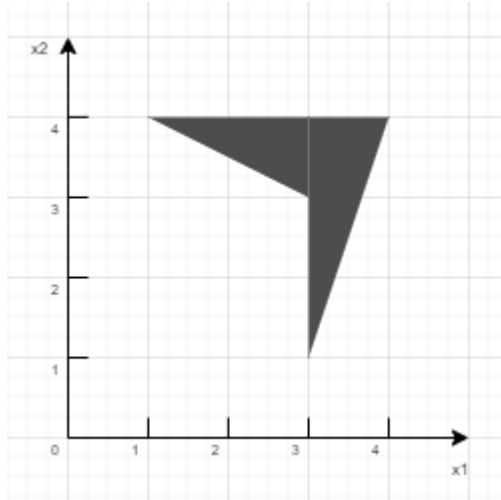
Artificial Neural Networks

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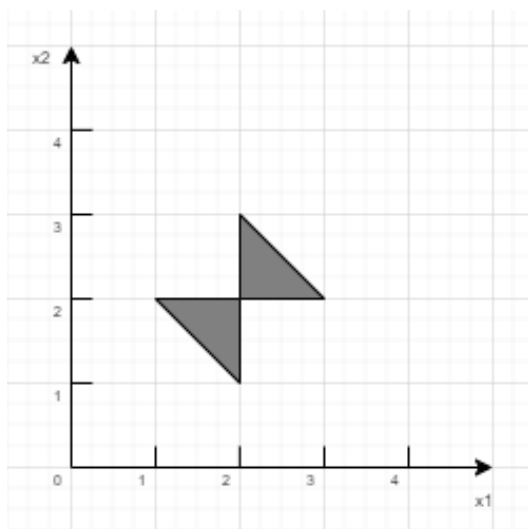
Task 1 – Threshold Logic Units

Create a 3-Layer Perceptron to Predict the Area Inside the Shape to +1 and outside to 0



Task 2 – Radial Basis Functions

The output should be 1 inside the shaded area and ≤ 0 outside the shaded area. The border doesn't matter. Each node can have different distances, but all distance functions should be from the Minkowski family. Provide all parameters, including centers, distances, input, activation, and respective weights.

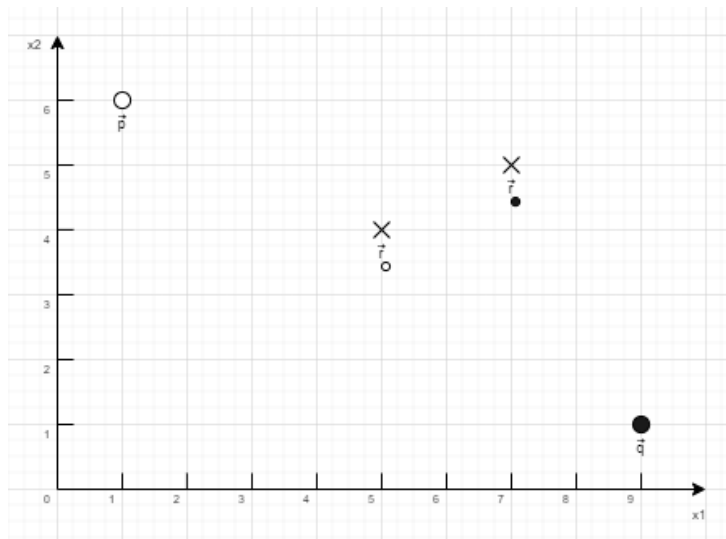


Task 3 – Function Approximation

- Approximate Function $y = x^2 + 2x + 2$ within the Range $[-4, 4]$ using a 3-Layer Perceptron of nodes $8 \leq n \leq 12$
- Approximate Function $y = x^2 + 2x + 2$ within the Range $[-4, 4]$ using a Radial Basis Function of nodes $8 \leq n \leq 12$
- How can the approximations be improved. Give atleast 2 Options

Task 4 – Learning Vector Quantization

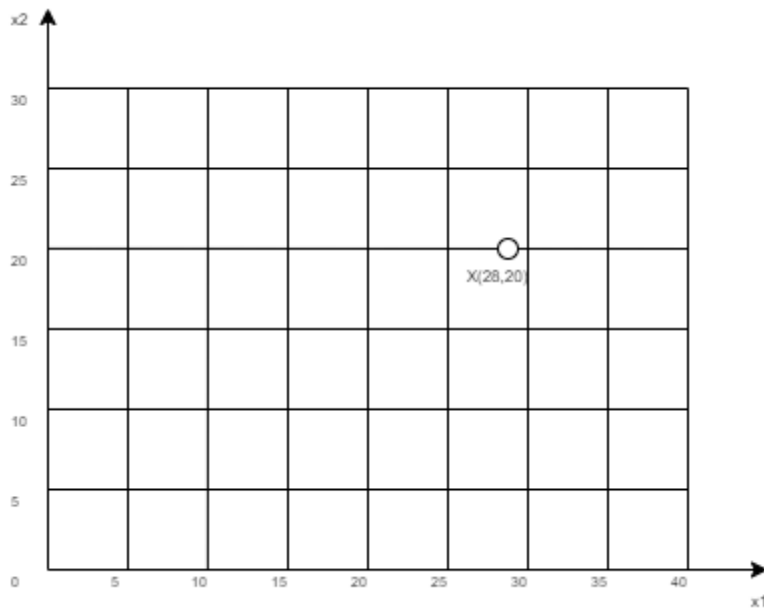
- For two classes, $\bigcirc$ and $\bullet$ and two data points, one for each class, compute the change in the codebook vector for one epoch with a learning rate of $\eta = 0.5$. The data point order is $\vec{p}, \vec{q}$
 - Class information ignored
 - Class information taken into account



b) Self-Organizing Maps

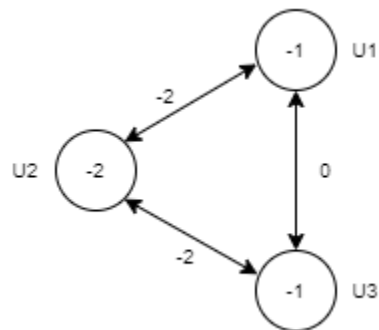
Compute the change in the codebook vectors for the given values $\vec{x} = (28, 20)$.

Use a learning rate $\eta = 0.5$ the Euclidean distance function, and Gaussian activation with a radius $\sigma_u = 2$



Task 5 – Hopfield Networks

Draw the complete State Graph. No Update order is Specified. Consider all possible Successor States



Task 6 – MCQ (mark True/False. Negative points for Wrong answers)

a. Types of Neural Networks

- Multi-Layer Perceptron
- Radial Basis Function Networks
- Learning Vector Quantization
- Self-Organizing Maps
- Hopfield Networks

b. Threshold Logic Units

- TLU with n inputs. n -dim hypercube.
- For 2 Inputs, there are More Linearly Separable Combinations than Linearly Non-Separable Combinations
- In a 2 Class Problem, if the 2 classes are Linearly Separable, there exist at least one point which is common to both Convex Hulls

c. Multi-Layer Perceptron and Radial Basis Functions

- MLPs (with linear activation functions) and RBFNs Compute Power comparison

d. Training Neural Networks

- Quick Prop
- RMS Prop and Adam

e. Distance Functions

- Triangular
- Associative
- Transitive
- Identity

f. Hopfield Networks

- Has Symmetric Weights with 1s on the Diagonal
- No of Hidden Neurons equal to no of Input Neurons
- Updating One Value and Going to a new state, if it's replaced immediately, it will go back to the previous state due to symmetry
- Bipolar and Unipolar equivalent in compute power
- Updating Nodes asynchronously without skipping will always lead to stability